

Renewable sources of energy often struggle to compete with other forms of power such as coal and natural gas. The two largest inefficiencies that renewable sources face are environmental effects on power generation and general maintenance costs.

Environmental Effects on Renewable Power Sources

Problem

Existing renewable energy sources lack efficiency during rainy conditions. Rain blocks sun from sites using focused mirrors, lowering their efficiency to near 0%, wind-turbines often have to be locked after a certain wind speed in order to avoid damage to the internals, and solar panels have their efficiency reduced to around 10-25% [10]. Solar panels, in particular, are susceptible to many environmental conditions such as: smog, dust, fog, clouds, rain, and dew. Additionally, due to the photovoltaic materials often used, solar panels tend to pool water on their surface. This creates a thin film that reduces the efficiency of the solar panel despite the source of the water missing [10].

Maintenance Requirements of Renewable Power Sources

Existing renewable sources often have lower operational efficiencies compared to other methods of power generation because of their need for maintenance. In this case, maintenance can be broken down into three parts: preventive, in order to stop any issues from occurring; corrective, to fix any issue that has occurred; and predictive, which uses data like wear marks and common modes of failure in order to apply preventive maintenance only when needed [4]. Between these types of maintenance, the average cost to maintain a small array of solar panels ranges from \$300 to \$700 annually with large amounts of inoperable downtime [5].

Research

Costs

Common piezoelectric materials such as Lead Zirconate Titanate and Polyvinylidene Fluoride can range in costs anywhere from \$50 to \$200 per kilogram depending on the quality and scale of the production processes [3]. The materials themselves are expensive, but the overall manufacturing into usable electrical-harvesting products makes up for a bulk of the price. Given that the average piezoelectric power output exists on scales of milliwatts to a few watts; it puts the cost-per-watt ratio of piezoelectrics higher than any common power generation method that is currently widely adopted [6]. These factors are offset by the overall reliability and independence of piezoelectric systems allowing for large amounts of power to be generated with little to no intervention or supervision. This allows for reasonable ROI (return on investment) for solar and mirror plants that do not need to invest in expensive specialist labor costs.

Power Output

Piezoelectric materials convert incurred mechanical (kinetic) energy into electrical energy through their electromechanical coupling properties.

The power produced by the piezo electric is directly related to the strain-force relation (S_{ij}) of the material [7].

$$S_{ij} = s_{ijkl} T_{kl} + d_{ijk} E_k \quad (\text{Eq. 1})$$

By taking the elastic compliance tensor s_{ijkl} in Pa^{-1} of the given material and multiplying it with the stress tensor T_{kl} in Pa being introduced to the material, you are given the overall mechanical offset of the equation. On the other side is the piezoelectric strain

coefficient d_{ijk} which is in units of C/N or m/V of the material; which determines how much strain is produced per a given unit provided by an electric field. The electric field vector E_k is then multiplied into the piezoelectric strain coefficient to determine the overall electric strain on the material.

The other important equation is for the electric displacement-field relation given as D_i (the charge over the total area of the piezoelectric) in [7]:

$$D_i = d_{ijk} T_{jk} + \epsilon_{ij} E_j \quad (\text{Eq. 2})$$

The piezoelectric coefficient d_{ijk} is based on the material and is multiplied into the mechanical stress T_{jk} being put into the material which is then added with the dielectric permittivity tensor ϵ_{ij} which is the coefficient based on the material that determines how easily the material polarizes in the presence of an electric field E_j .

Simplifying these two equations into a single dimension would give:

$$S = s^E T + dE \quad (\text{Eq. 3a})$$

$$D = dT + \epsilon^T E \quad (\text{Eq. 3b})$$

Solving for E using 3a we get and plugging that into the simplified displacement-field relation equation:

$$E = \frac{S-s^E T}{d} \quad (\text{Eq. 4a})$$

$$D = \frac{\epsilon^T}{d} S + (d - \frac{\epsilon^T s^E}{d}) T \quad (\text{Eq. 4b})$$

Now it is possible to solve for both voltage and current given the area A of the piezoelectric [7]:

$$D = \frac{Q}{A} \Rightarrow I = A \frac{dD}{dt} \quad (\text{Eq. 5a})$$

$$V = E * t \quad (\text{Eq. 5b})$$

So the final function to determine the power output of a given piezoelectric surface ,where t_h is the thickness of the material, is [7]:

$$P(t) = (\frac{S(t)-s^E T(t)}{d} * t_h) * A * \frac{d}{dt} [\frac{S(t)\epsilon^T}{d} + (d - \frac{\epsilon^T s^E}{d}) T(t)] \quad (\text{Eq. 6})$$

Practically, this means that the higher the effective strain, the larger the area, and the thicker the material: the higher the power output. Given typical values of modern piezoelectrics the graph would look similar to Fig. 1.

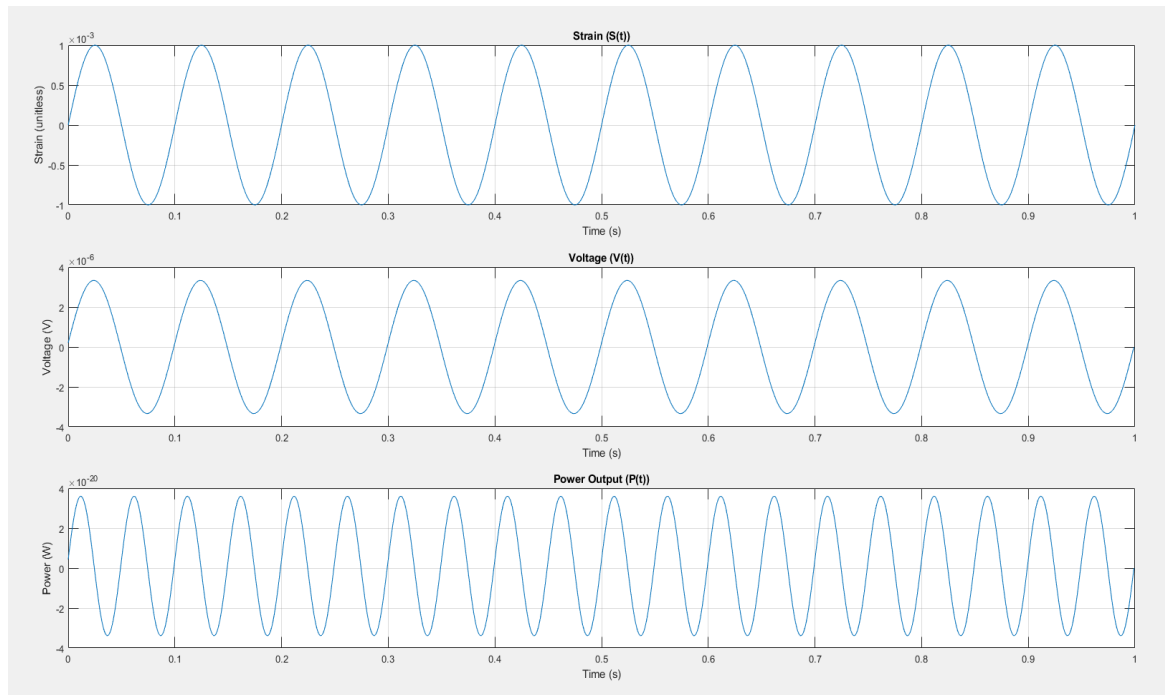


Fig. 1 Graph of Strain, Voltage, and Power output of an example piezoelectric material

Scalability

One of the biggest strengths of renewable energy is its ability to scale to increase energy efficiency. Unfortunately, at the moment it is not very feasible for piezoelectric energy generators to be implemented into large-scale harvesting. The main issue comes from their relatively limited energy outputs individually, meaning that adding more and more doesn't have enough benefit to counteract its cost. However, research into amplification of energy output and certain specific use cases for low power devices utilizing multilayered piezoelectric generators shows much promise for the future of piezoelectric scalability [2].

Environmental Effect

One issue that piezoelectrics have is their high initial energy cost. Piezoelectric materials are mostly ceramic and as such need to be heated in furnaces. The power these furnaces require to create these ceramics is around 10kW, while the individual energy

harvesters produce around μW to mW . With these furnaces producing 4000 to 5000 energy harvesters, the best case scenario would have the energy cost being offset by the produced energy in half a year, and worse case hundreds of years. The best way to avoid this problem would be to avoid high temperature furnaces to avoid high energy requirements to produce energy [10].

Another issue comes from piezoelectronics containing lead, which can be toxic for the environment. Normally, this would simply be designed out, considering there already exist regulations against lead in electronics. However, the reality is that the more prominent lead-free designs are simply not as energy efficient as those containing lead, so it would be less efficient to be more environmentally friendly. More research needs to be done in order to find a suitable solution to this problem, which currently has no good solution [10].

Solution

Piezoelectric systems provide a potential solution to generating renewable energy when otherwise not available. Unlike solar panels, which rely on sunny weather, such systems can operate during storms by utilizing mechanical energy provided by the rain [8]. When deployed on rooftops in tandem with solar panels, households can generate electricity regardless of whether the sun is out or not. By combining a piezoelectric generator with some sort of machine that oscillates pressure, or any system where there is a predictable change in pressure, power can be generated in rainy weather. On a large scale, such as multiple rooftops on a rainy day, there can be a significant amount of potential and kinetic energy in the water that can be harnessed in this way. Similar pressure-based systems have already been implemented in countries such as Japan, where footsteps power on lights in high traffic areas.

Despite the initial cost in the raw materials and production of these systems, they are low maintenance. Current renewable sources of electricity, such as solar panels and wind turbines, require maintenance as they get exposed to weather for long periods of time [1].

However, due to their nature, weather does not subject piezoelectric systems to the same wear and tear as they do not contain mechanical parts and also do not get inhibited by hard water deposits. This form of preventative maintenance guarantees a long operational lifespan, reducing the need for extra labour as well as providing trust that the system is sustainable. Piezoelectric headphones made in the early 1900s, for example, still work well.

Moreover, piezoelectric systems can function in parallel with the existing energy infrastructure. Using such systems would not require a major overhaul in the current system. Because of the infrastructure built for other renewable energy sources such as solar panels and windmill generators, we already have the basis for how to store the energy that would be captured by these piezoelectric systems. Because many of these systems are still being built, they are built with the expectation that they will be expanded for more renewable energy options, making it much easier to implement the piezoelectric solution.

Because of their lack of maintenance, seamless integration, and reliability, piezoelectric systems find a niche for use in remote areas [1]. They have the ability to power devices such as lights in places that may not be connected to an existing electrical grid. This differs from solar panels and wind turbines, which require both specialized maintenance and extra labor.

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